

Assignment 3.1

GitHub Project to EC2 Deployment using Jenkins Pipeline

Objective

Deploy a website from the GitHub repository Shantanumajan6/project to an AWS EC2 Linux VM using a Jenkins Pipeline job. The pipeline checks out the repository, installs and starts Apache HTTP Server if required, copies the project files to /var/www/html, and verifies that the website is accessible through the EC2 public IP.

Difference from Assignment 3: Assignment 3 used a Jenkins Freestyle project with an Execute shell build step. Assignment 3.1 performs the same deployment with a Jenkins Pipeline, where each task is a separate stage.

Architecture / Flow

GitHub Repository → Jenkins Pipeline Job → Checkout Stage → Jenkins Workspace → Install/Start Apache Stage → Deploy Website Stage → /var/www/html → Verify Stage → Apache :80 → Browser

Item	Value
Source	GitHub repository
Repository	https://github.com/Shantanumajan6/project
Branch	main
Jenkins Job	Project-Pipeline-Deployment
Jenkins Type	Pipeline (Pipeline script)
Destination	AWS EC2 Linux VM
Web Server	Apache HTTP Server (httpd)
Document Root	/var/www/html
HTTP Port	80

Jenkins Pipeline Deployment Steps

1. Verify the Linux VM

Connect to the EC2 VM using SSH.

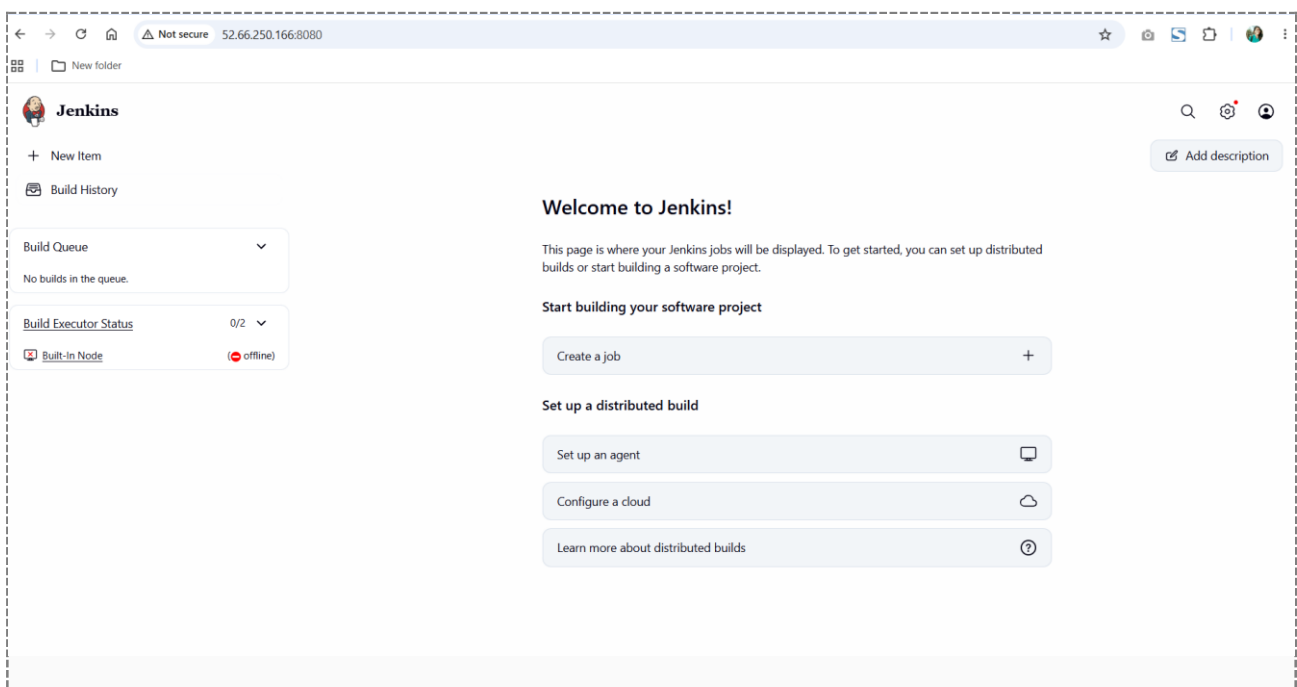
```
ssh -i "key_pair.pem" ec2-user@ec2-52-66-250-166.ap-south-1.compute.amazonaws.com
```

Note: Apache installation and service management will be performed by the Jenkins Pipeline.

[illegible]

2. Verify Jenkins

Verify Jenkins is running and open the Jenkins dashboard in the browser (<http://52.66.250.166:8080/>).



3. Install and Verify Git

```
yum install git -y
git --version
```

Git must be available on the Jenkins VM so that Jenkins can check out the GitHub repository.

```
[root@ip-172-31-1-51 ~]# git --version
git version 2.50.1
[root@ip-172-31-1-51 ~]#
```

4. Verify Pipeline Plugin

Go to: Manage Jenkins → Plugins → Installed plugins. Confirm the Pipeline and Git plugins are installed. If not, install them and restart Jenkins.

5. Create Jenkins Pipeline Project

In Jenkins:

1. Click New Item
2. Enter: Assignment3.1-Pipeline
3. Select Pipeline
4. Click OK

Jenkins / All / New Item

New Item

Enter an item name

Assignment3.1-Pipeline

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
- Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
- Duplicate an existing item
Create a new item by copying the configuration of an existing one.

OK

5. 6.

6. Configure the Pipeline

In the Pipeline section of the job configuration:

1. Definition: Pipeline script
2. Script: paste the pipeline script given below
3. Click Save

Use the following Declarative Pipeline script:

```
pipeline {
  agent any

  stages {
    stage('Checkout') {
      steps {
        git branch: 'main',
          url: 'https://github.com/Shantanumajan6/project'
      }
    }
  }
}
```

```

stage('Install Apache') {
    steps {
        sh '''
        echo "==== Installing Apache ====="
        sudo yum install -y httpd
        sudo systemctl enable httpd
        sudo systemctl start httpd
        '''
    }
}

stage('Deploy Website') {
    steps {
        sh '''
        echo "==== Deploying Website ====="
        sudo cp -r ./* /var/www/html/
        sudo chown -R apache:apache /var/www/html
        sudo chmod -R 755 /var/www/html
        ls -la /var/www/html/
        '''
    }
}

stage('Verify Deployment') {
    steps {
        sh '''
        echo "==== Apache Status ====="
        sudo systemctl status httpd --no-pager
        echo "==== Testing Website ====="
        curl -I http://localhost
        '''
    }
}
}

```

Stages: Checkout → Install Apache → Deploy Website → Verify Deployment.

Note: Manual git clone is not required. The Checkout stage performs the Git checkout from <https://github.com/Shantanumajan6/project> (branch main). Use whichever branch holds the code, for example master.

7. Save and Build

Click Save, then Build Now. Jenkins will check out the repository, install Apache, copy the files, and verify the deployment. Open the build and check the Stage View to confirm every stage is green.

Open the build and select Console Output. Confirm the build ends with Finished: SUCCESS.

Jenkins

Project-Freestyle-Deploy...

#1

Status

Changes

Console Output

Delete build #1

Git Build Data

Edit build information

Download

Copy

View as plain text

Console

```

Started by user Adelin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Project-Freestyle-Deployment
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/Shantanujana6/project
> git init /var/lib/jenkins/workspace/Project-Freestyle-Deployment # timeout=10
Fetching upstream changes from https://github.com/Shantanujana6/project
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/Shantanujana6/project +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/Shantanujana6/project # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision 4e0e5cf4feb2e7344c3d80594615248b9eb69 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f 4e0e5cf4feb2e7344c3d80594615248b9eb69 # timeout=10
Commit message: "Update web.xml"
First time build. Skipping changelog.
[Project-Freestyle-Deployment] $ /bin/bash /tmp/jenkins10735130050513243604.sh
===== Starting Deployment =====
===== Jenkins Workspace =====
/var/lib/jenkins/workspace/Project-Freestyle-Deployment
total 32
drwxr-xr-x. 6 jenkins jenkins 120 Oct 3 08:16 .
drwxr-xr-x. 10 jenkins jenkins 16384 Oct 3 08:16 ..
-rw-r--r--. 1 jenkins jenkins 1023 Oct 3 08:16 .classpath
drwxr-xr-x. 7 jenkins jenkins 146 Oct 3 08:16 .git
drwxr-xr-x. 3 jenkins jenkins 23 Oct 3 08:16 .github
-rw-r--r--. 1 jenkins jenkins 861 Oct 3 08:16 .project
drwxr-xr-x. 2 jenkins jenkins 133 Oct 3 08:16 .settings
-rw-r--r--. 1 jenkins jenkins 507 Oct 3 08:16 Jenkinsfile
-rw-r--r--. 1 jenkins jenkins 949 Oct 3 08:16 pom.xml
drwxr-xr-x. 3 jenkins jenkins 18 Oct 3 08:16 src
===== Installing Apache =====
Last metadata expiration check: 5:10:46 ago on Sat Oct 3 03:05:55 2026.
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
===== Starting Apache =====
===== Deploying Website =====
===== Deployed Files =====
total 16
drwxr-xr-x. 6 apache apache 115 Oct 3 08:16 .
drwxr-xr-x. 4 root root 33 Oct 3 03:12 ..
-rwxr-xr-x. 1 apache apache 507 Oct 3 08:16 Jenkinsfile
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R1
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R2
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R3
-rwxr-xr-x. 1 apache apache 27 Oct 3 07:23 dev.html
-rwxr-xr-x. 1 apache apache 38 Oct 3 07:23 index.html
-rwxr-xr-x. 1 apache apache 949 Oct 3 08:16 pom.xml
drwxr-xr-x. 3 apache apache 18 Oct 3 08:16 src
===== Apache Status =====
* httpd.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
Active: active (running) since Sat 2026-10-03 03:14:37 UTC; 5h 2min ago
Docs: man:httpd.service(8)
Main PID: 28835 (httpd)
Status: "Total requests: 38; Idle/Busy workers 100/0; Requests/sec: 0.0021; Bytes served/sec: 2 B/sec"
Tasks: 230 (limit: 1059)
Memory: 18.8M
CPU: 20.591s
CGroup: /system.slice/httpd.service
└─28835 /usr/sbin/httpd -DFOREGROUND
└─28858 /usr/sbin/httpd -DFOREGROUND
└─28860 /usr/sbin/httpd -DFOREGROUND
└─28861 /usr/sbin/httpd -DFOREGROUND
└─28869 /usr/sbin/httpd -DFOREGROUND
└─29135 /usr/sbin/httpd -DFOREGROUND

Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal httpd[28835]: Server configured, listening on: port 80
===== Testing Website =====
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed

  0   0   0    0  0  0     0   0  0      0  0  0   0
  0  30   0    0  0  0     0   0  0      0  0  0   0
  0  30   0    0  0  0     0   0  0      0  0  0   0
  0  20   0    0  0  0     0   0  0      0  0  0   0

HTTP/1.1 200 OK
Date: Sat, 03 Oct 2026 08:16:43 GMT
Server: Apache/2.4.68 (Amazon Linux)
Last-Modified: Sat, 03 Oct 2026 07:23:20 GMT
ETag: "1e-65cea5f87af7"
Accept-Ranges: bytes
Content-Length: 30
Content-Type: text/html; charset=UTF-8

===== Deployment Completed =====
Finished: SUCCESS

```

Jenkins 2.580.1

One Jenkins pipeline run demonstrates the full flow: Git checkout → Apache installation → Apache startup → website deployment → permissions → verification.

8. Verify Deployment on EC2 VM

After a successful pipeline run, SSH into the VM and run:

```
ls -la /var/www/html/
```

Expected: the project files from the repository (for example index.html).

Then:

```
curl -I http://localhost:8081/LoginWebApp/
```

Expected: HTTP/1.1 200 OK

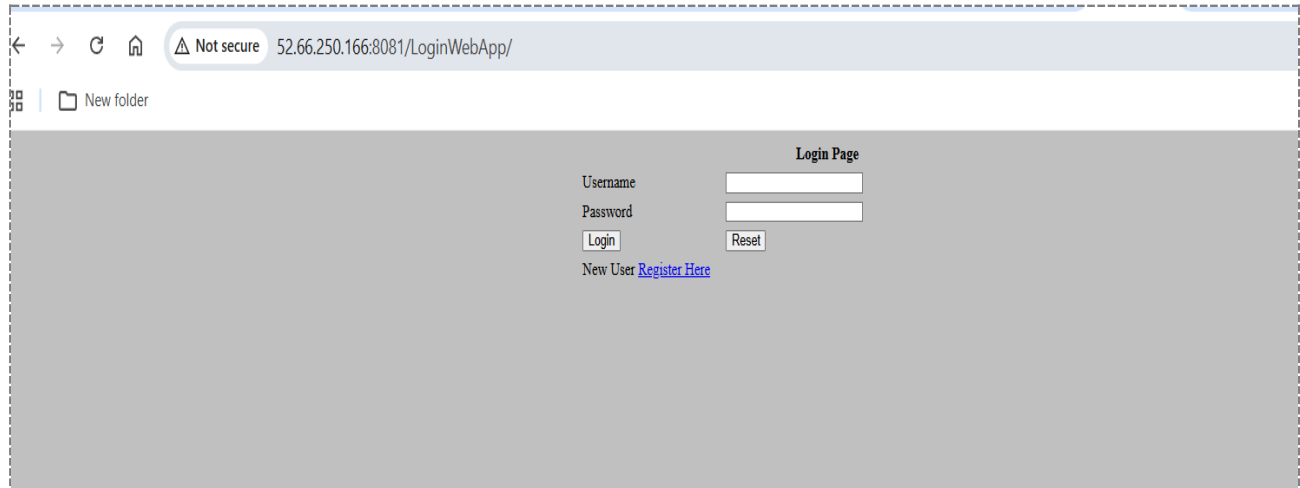
```
[root@ip-172-31-1-51 Assignment3.1-Pipeline]# curl -I http://localhost:8081/LoginWebApp/
HTTP/1.1 200
Set-Cookie: JSESSIONID=43DC8C1990D992C1E0C2F98EC44F6CFA; Path=/LoginWebApp; HttpOnly
Content-Type: text/html; charset=UTF-8
Content-Length: 1286
Date: Sat, 03 Oct 2026 11:09:26 GMT

[root@ip-172-31-1-51 Assignment3.1-Pipeline]#
```

9. Verify Website from Browser

Open: <http://52.66.250.166/index.html>

Note: Make sure port 80 is open in the EC2 Security Group (inbound rule: HTTP, TCP 80).



Outcome

- Pipeline job pulls code from the GitHub repository.
- Apache is installed and started automatically.
- Website files are deployed to /var/www/html.
- The website is accessible via the EC2 public IP.